

# NEUROLINK WEAR



# NeuroLink Wear

## Smart Wearable for Predictive Health Prediction & Detection

### Project Overview

NeuroLink Wear is an AI-powered smart wearable system designed to monitor vital signs and predict early signs of stress or illness using IoT and machine learning. The system integrates anomaly detection models and Large Language Models (LLMs) to provide real-time alerts and personalized recommendations.

### Core Sensors

| Sensor         | Function                           |
|----------------|------------------------------------|
| MAX30102 (PPG) | Heart Rate, SpO <sub>2</sub> , HRV |
| GSR            | Stress / Skin Conductance          |
| MPU6050        | Motion, Tremor, Fall Detection     |
| MLX90614       | Skin Temperature                   |
| NEO-6M         | GPS Location                       |

### System Architecture

#### 1. Edge Layer (ESP32)

- Collects sensor data (HR, HRV, GSR, temperature, motion)
- Performs feature extraction (RMSSD, variance, trends)
- Applies lightweight threshold filtering
- Sends data via MQTT every 1–5 seconds

#### 2. AI Core Layer

##### Anomaly Detection Engine

- Isolation Forest (primary model)
- Statistical threshold rules (fallback)
- Optional: Autoencoder / LSTM for time-series analysis

##### LLM Layer

- Interprets anomalies
- Generates explanations
- Provides personalized recommendations

### 3. User Application

- Displays real-time vitals
- Sends alerts (stress, fever, low oxygen)
- Provides actionable advice



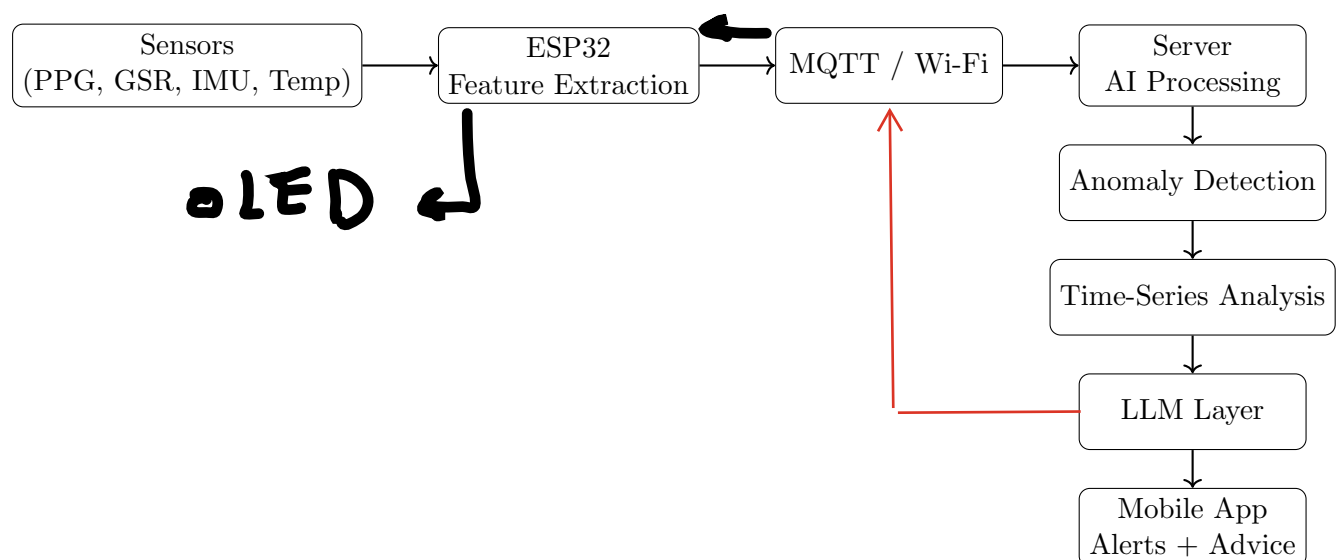
### Anomaly Detection Logic

| Condition    | Indicators           | Detection Logic               |
|--------------|----------------------|-------------------------------|
| Stress       | HRV ↓, GSR ↑, Temp ↑ | Combined anomaly score        |
| Panic Attack | HR ↑↑, GSR spike     | Sudden deviation + no motion  |
| Fever        | Temp ↑ trend         | Continuous increase over time |
| Low Oxygen   | SpO <sub>2</sub> ↓   | Threshold-based detection     |
| Fatigue      | HRV low              | Long-term trend analysis      |
| Fall         | IMU spike            | Sudden acceleration           |

### AI Multi-Agent System (Optional)

| Agent           | Role                      |
|-----------------|---------------------------|
| Detection Agent | Runs anomaly models       |
| Health Agent    | Interprets condition      |
| Advice Agent    | Generates recommendations |
| Alert Agent     | Sends notifications       |

### Data Pipeline Architecture



## Key Features

- Real-time anomaly detection
- Time-series prediction of health conditions
- HRV-based stress detection
- LLM-powered personalized advice
- Energy-efficient edge computing

## Conclusion

NeuroLink Wear integrates IoT sensing, anomaly detection, time-series modeling, and LLM-based intelligence to create a next-generation health monitoring system capable of predicting, explaining, and guiding user health in real time.